

Problem 1

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Problem 1 Find the most general antiderivative of the function

$$g(t) = e^{-2t} - 5 + 6\sqrt{t} - \frac{7}{t} + \frac{5}{1+t^2}$$

Explanation. The family of all antiderivatives of g is given by

$$\int \left(e^{-2t} - 5 + 6\sqrt{t} - \frac{7}{t} + \frac{5}{1+t^2} \right) dt.$$

We can apply the Sum Rule and Constant Multiple Rule.

$$\int \left(e^{-2t} - 5 + 6\sqrt{t} - \frac{7}{t} + \frac{5}{1+t^2} \right) dt = \int e^{-2t} dt - 5 \int dt + 6 \int \sqrt{t} dt - 7 \int \frac{1}{t} dt + 5 \int \frac{1}{1+t^2} dt$$

And finally,

$$\int \left(e^{-2t} - 5 + 6\sqrt{t} - \frac{7}{t} + \frac{5}{1+t^2} \right) dt = -\frac{1}{2}e^{-2t} - 5t + 4t^{\frac{3}{2}} - 7 \ln |t| + 5 \arctan t + C.$$